

Monkey business

Is it worth tracking benchmark indices such as the FTSE 100, or would an alternative index created randomly by monkeys produce better returns? A Cass team found some surprising answers, writes Simoney Girard.

Most people are aware of the so-called infinite monkey theorem, which states that a monkey hitting keys at random on a typewriter keyboard for an infinite amount of time will eventually replicate the entire works of William Shakespeare.

Mathematicians thrive on statistical theories such as this and gamblers hope that random combinations of numbers will enable them to live in luxury. But could random combinations not only achieve a stated goal, but do so almost ten million times a year for 43 years?

The answer is yes, according to research by Andrew Clare, Professor of Asset Management, Dr Nick Motson, Lecturer in Finance, and Steve Thomas, Professor of Finance, a Cass team that set out to find whether investments following randomly constructed indices could outperform traditional market-cap weighted indices such as the FTSE 100, and whether investors could be best served by using nontraditional indices. The study*, commissioned by the consultancy Aon Hewitt, comprised two series of experiments: part one focused on heuristic and optimised weighting schemes, and part two focused on fundamental weighting schemes.

Most stock market indices around the world are constructed according to market capitalisation; that is, their larger components carry a larger percentage weighting. It is the simplicity of these indices that has made them so popular with investors.

Passive tracking

The research team set out to establish whether passively tracking a market-cap index did indeed produce the highest possible returns.

A sample of 1,000 companies with traded equities was used. Professor Clare says: "We programmed a computer to randomly pick and weight each of the 1,000 stocks in the sample; we effectively simulated the stock-picking abilities of a monkey. The process was repeated ten million times over each of the 43 years of the study [1968-2011].

"The results of this experiment showed that many of the monkey fund managers would have generated a superior performance than was produced by some of the alternative indexing techniques. However, perhaps most shockingly, we found that nearly every one of the ten million monkey fund managers beat the performance of the market-cap weighted index."

Simian benchmark

He adds: "One of the implications of our work is that we should perhaps be benchmarking our fund managers against monkeys rather than against a cap-weighted index!"

From the sample of 1,000, each randomly chosen stock would be allocated a 0.1 per cent weighting into a portfolio. This was repeated 1,000 times, to create a 100 per cent invested "index".

A stock could get picked more than once but, according to Professor Clare, it was "highly unlikely" that a single stock would be picked so many times within one index construction that it could hold a 10 per cent or 15 per cent weighting.

He says: "The probability of creating the sort of concentration by chance that an investor would get by investing in a market-cap weighted index like the S&P500, is extremely small."

The Cass team then tested a host of alternative indexing strategies. One was to weight stocks equally. Instead of, say, the FTSE 100 having an 8 per cent weighting in BP, each stock was given equal importance.

Another was to take the stocks with the lowest overall price fluctuation and give these a proportionately higher weighting in the index. This is called inverse volatility.

Measuring risk

The team also constructed indices using optimisation techniques such as inputting certain variables. This would provide an expected return given certain parameters, such as risk measurements, or diversification based on a Sharpe ratio (an indication of how much the return on each stock can be attributed to risk).

The team weighted stocks based on fundamental company information other than its share price and determined the index weights of US equities every year from 1968 to 2011. For this, four alternative measures of company size, or scale, were used: total annual dividend; total annual cashflow; net asset value and total annual sales.

According to the report: "Each of these criteria produces a different weight for each company and, therefore, a different index. For example, when we use total dividends as the indicator of company scale, the company that made the largest dividend payment over a particular period would have the highest weight in the index, while the company that paid out the smallest dividend would have the least weight."

The team applied the same criteria to the other measures of scale to produce four fundamentally weighted indices. Would the monkeys generate a superior performance?

Superior performance

"The results were incredible," says co-author Dr Nick Motson. "The experiment showed that many of the monkey index constructors would have generated a superior performance than was produced by some of the alternative indexing techniques.

"However, perhaps most shockingly, we found that nearly every one of the ten million monkey fund managers beat the performance of the market-cap weighted index.

"It is almost impossible for the monkeys to come up with a set of weights that matched the real market-cap weighted index. So we knew that an investor would not come across a market-cap weighted portfolio by pure chance.

"What we hadn't known, what hadn't been certain, was that the performance of a randomly generated index would beat the market-cap index. We were expecting the market cap index to beat at least some of them. It beat hardly any of the ten million monkeys over the 43-year period." The study confirmed what Aon Hewitt's pension

consultants had long believed: that investors - institutional or retail - who unwaveringly follow a market-cap weighted benchmark index in the hope of generating good returns, are doing themselves a disservice.

Yesterday's winners

Dr Motson says: "One of the criticisms of market-cap weighting is that investors are buying yesterday's winners, despite the industry repeating the mantra 'Past performance is not a guide for future performance'."

John Belgrove, senior partner at Aon Hewitt, believes the "consistent academic rigour" of this research will help investors to understand better the opportunities and risks available in "smart beta" funds, which create their own indices based on variables such as volatility, value or dividend and track them.

He says: "This work sheds fresh light on the age-old active/passive industry debate. Inherent weaknesses in cap-weighted investment strategies are well documented, although they have been an enduring and challenging benchmark for active managers to beat."

Even though alternative ways of index investing have been shown to offer lower risks and higher rewards, the product range is still considered esoteric. Institutional players and trustees are suspicious of change, and the retail market is a generation behind, according to investment advisors.

Default option

Jason Butler, a director at London based Bloomsbury Financial Planning and author of The Financial Times Guide to Wealth Management, says: "We have been using alternative ways of providing passive investment strategies for our clients for years.

"There are many ways to track an index, such as using tilts towards higher dividends or value stocks. Market cap is the default option and with this option, market dynamics also dampen performance further by charging on the spread or forcing you to buy initial public offerings.

"There are far more effective, efficient and intelligent ways of getting passive strategies - but so far it is an evolution that is yet to happen. Hopefully the Cass study will help prise the door open."